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Portable fruit picker with adjustable height

Abstract

A portable fruit picker with an adjustable height comprises a picking module and a container part, wherein the picking module comprises a picking part and a supporting part, and the picking part comprises a collecting part, a grabbing part and a conveying part which are engaged with the collecting part; the conveying part has elastic characteristics; the supporting part comprises a plurality of detachably connected supporting bodies; the container part is provided with a fixing part. The user can adjust the height of the supporting part independently through the detachably connected supporting bodies, and the container is provided with the fixing part, so that the user can fix the container on himself.

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Background/Summary**TECHNICAL FIELD**

(1) The present invention relates to the field of high-altitude goods collection, in particular to a portable fruit picker with an adjustable height.

BACKGROUND

(2) When fruit farmers are picking fruits, they usually climb to the trees to pick the high fruits, which is often accompanied by danger; And some high fruits can't be picked even if they climb to the tree. At this time, people may choose to beat them with long poles or leave them on the tree. However, if a long pole is used, the fruit may be destroyed, or it may be smashed when it falls to the ground; this will cause economic losses to fruit farmers.

(3) In U.S. Pat. No. 31,961,972, a manual fruit picking basket is provided, which can be supported by a slender hand-held rod for continuously picking fruits from trees without a ladder or lowering the picking basket. The picking basket is basically cylindrical and includes an open upper end for receiving fruits, a lower end which defines a handle engaging socket; the handle engaging socket is

suitable for receiving one end of an elongated hand-held rod for picking fruits high on a tree, and a central part defined between the upper end and the lower end, in which a position containing fruits is defined, the open upper end is defined by an edge, the upper end has a front part and a rear part; the front part comprises a basically open edge part, and the rear part comprises a plurality of spaced, elongated fruit engaging fingers, and the fingers extend upward beyond the edge, and have a hook-shaped structure so as to at least partially surround the fruit to be picked; an outlet located under the central part and fingers, wherein the outlet is located near the handle engaging socket but higher than the socket, and its diameter is such that fruits separated from fruit trees can pass through the outlet; and a slender tubular fruit conveying device suitable for being connected to a basket near the outlet, wherein the conveying device is made of elastic fabric, and the diameter of the outlet is usually smaller than that of the fruit to be picked, but the fruit can be stretched when it passes through the outlet from the central part of the basket and enters the conveying device; in use, the picking basket is usually placed at a certain angle with the ground, and the rear part of the upper end of the basket is closest to the ground, so that the fruit separated from the tree by fingers can move down along the fingers, move along the inside of the central part of the basket, then immediately roll to the outlet device and then move down along the conveying device.

(4) However, the height of the hand-held pole supporting the picking basket cannot be adjusted, and when users can use the picking basket to pick fruits with different heights, it will cause inconvenience if the rod body is too long or not long enough. Its tubular fruit conveying device is made of elastic fabric, which can be stretched horizontally but not longitudinally, which makes it possible for the conveying device to produce a U-shaped structure after picking from a high place to reduce the height, so that the fruits are piled up in the conveying device; the fruit container is placed at the outlet of the conveying device, but the fruit container is placed on the ground, and there is no fixation between the conveying device and the container, which is easy to make the fruit fall out of the container from the outlet of the conveying device; the fruit container is placed on the ground, and it may need to be moved constantly when picking fruits, which leads to a bad picking experience.

SUMMARY

(5) The terms “invention,” “the invention,” “this invention” and “the present invention” used in this patent are intended to refer broadly to all of the subject matter of this patent and the patent claims below. Statements containing these terms should be understood not to limit the subject matter described herein or to limit the meaning or scope of the patent claims below. Embodiments of the invention covered by this patent are defined by the claims below, not this summary. This summary is a high-level overview of various embodiments of the invention and introduces some of the concepts that are further described in the Detailed Description section below. This summary is not intended to identify key or essential features of the claimed subject matter, nor is it intended to be used in isolation to determine the scope of the claimed subject matter. The subject matter should be understood by reference to appropriate portions of the entire specification of this patent, any or all drawings and each claim.

(6) In order to solve the problems that the height of the supporting part rod is not adjustable and the container is inconvenient to move, the present invention provides a portable fruit picker with an adjustable height.

(7) In order to achieve the object of height adjustment and portability, the technical solution of the present invention is as follows:

(8) A portable fruit picker with an adjustable height includes a picking module and a container part, wherein the picking module is detachably connected with the container part; the picking module includes a picking part and a supporting part, and the picking part is detachably connected with the supporting part; the picking part includes a collecting part, a grabbing part and a conveying part which are engaged with the collecting part; the collecting part is provided with a collecting inlet end and a collecting outlet end, the conveying part is provided with an input port end and an output

port end, the grabbing part is engaged with the collecting inlet end, and the input port end of the conveying part is detachably connected with the collecting outlet end of the collecting part; the conveying part has elastic characteristics, so that the conveying part can stretch longitudinally.

(9) The supporting part includes a first supporting body and at least one second supporting body, and the first supporting body and the second supporting body are detachably connected, the first supporting body has a holding end and a first connecting end, one end of the second supporting body is engaged with the connecting end of the first supporting body, and the other end of the second supporting body is detachably connected with the picking part; the container part is provided with a first opening and a second opening, and the first opening is detachably connected with the output port end of the conveying part, the container part is provided with a fixing part, which includes a first fixing part and a second fixing part, and the second fixing part includes a first fixture and a second fixture, and the first fixture and the second fixture are detachably connected.

(10) In this solution, the user can adjust the height of the supporting part independently according to the use scene through the first supporting body and a plurality of second supporting bodies which are detachably connected; elastic materials are sewn in the conveying part, so that the conveying part can longitudinally extend and contract, and the length of the conveying part will be correspondingly shortened when the user picks from a high place to reduce the height, so as to avoid a U-shaped structure and prevent fruits from being accumulated in the conveying part; the conveying part is detachably connected with the container part, and the container part is provided with a fixing part, so that the user can fix the container part on himself, which is convenient for the user to change the picking position at any time.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) In order to explain the technical solution of this application more clearly, the drawings needed in the implementation will be briefly introduced below. Obviously, the drawings described below are only some implementations of this application. For those skilled in the art, other drawings can be obtained according to these drawings without creative work.
- (2) FIG. 1 is a schematic diagram of the overall structure of a fruit picker;
- (3) FIG. 2 is a schematic structural diagram of the picking part;
- (4) FIG. 3 is a sectional view taken at A-A in FIG. 2;
- (5) FIG. 4 is an enlarged view at c in FIG. 2;
- (6) FIG. 5 is a schematic structural view of the conveying part;
- (7) FIG. 6 is a schematic structural view of the first supporting body;
- (8) FIG. 7 is a schematic diagram of the structure of the connector;
- (9) FIG. 8 is a schematic structural view of the second supporting body;
- (10) FIG. 9 is a sectional view taken at B-B in FIG. 7;
- (11) FIG. 10 is a schematic view of the structure of the connecting sleeve;
- (12) FIG. 11 is a schematic structural view of the container part;
- (13) FIG. 12 is a schematic structural view of the fourth connecting part.
- (14) In the drawings: **1**, Picking part; **11**, Collecting part; **12**, Conveying part; **13**, Grabbing part; **111**, Collecting inlet; **112**, Collecting outlet; **113**, Stabilizer; **114**, Fifth connecting part; **121**, Input port end; **122**, Output port end; **123**, Buffer part; **131**, Strip-shaped part; **132**, Bending part; **133**, Grabbing gap; **2**, Supporting part; **21**, First supporting body; **22**, Second supporting body; **23**, Connector; **24**, Connecting sleeve; **211**, Holding end; **212**, First connecting end; **213**, Connecting part; **214**, First limiting inclined plane; **215**, First jack; **2111**, Groove; **2112**, Suspension part; **2113**, Hanging hole; **221**, First inserting end; **222**, Second connecting end; **223**, Third connecting part; **224**, Second limiting inclined plane; **225**, Second jack; **2211**, First connecting channel; **2212**,

Second connecting channel; **23**, Connector; **231**, Inserting end; **232**, Connecting end; **233**, First limiting part; **2311**, Guide inclined plane; **2321**, First connecting part; **2322**, Second connecting part; **24**, Connecting sleeve; **241**, Second limiting part; **3**, Container part; **31**, First fixing part; **32**, Second fixing part; **33**, First opening; **34**, Second opening; **35**, Fourth connecting part; **36**, Fixture; **37**, Third opening; **38**, Cover; **311**, Fixing hole; **321**, First fixture; **322**, Second fixture; **351**, Fixing groove.

DESCRIPTION OF EMBODIMENTS

(15) In describing the preferred embodiments, specific terminology will be resorted to for the sake of clarity. It is to be understood that each specific term includes all technical equivalents which operate in a similar manner to accomplish a similar purpose.

(16) While various aspects and features of certain embodiments have been summarized above, the following detailed description illustrates a few exemplary embodiments in further detail to enable one skilled in the art to practice such embodiments. Reference will now be made in detail to embodiments of the inventive concept, examples of which are illustrated in the accompanying drawings. The accompanying drawings are not necessarily drawn to scale. The described examples are provided for illustrative purposes and are not intended to limit the scope of the invention. It should be understood, however, that persons having ordinary skill in the art may practice the inventive concept without these specific details.

(17) It will be understood that, although the terms first, second, etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first attachment could be termed a second attachment, and, similarly, a second attachment could be termed a first attachment, without departing from the scope of the inventive concept.

(18) It will be understood that when an element or layer is referred to as being “on,” “coupled to,” or “connected to” another element or layer, it can be directly on, directly coupled to or directly connected to the other element or layer, or intervening elements or layers may be present. In contrast, when an element is referred to as being “directly on,” “directly coupled to,” or “directly connected to” another element or layer, there are no intervening elements or layers present. Like numbers refer to like elements throughout. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

(19) As used in the description of the inventive concept and the appended claims, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates other.

(20) As shown in FIGS. **1** to **4**, a portable fruit picker with an adjustable height includes a picking module and a container part **3**, wherein the picking module is detachably connected with the container part **3**; the picking module comprises a picking part **1** and a supporting part **2**, wherein the picking part **1** comprises a collecting part **11**, and a grabbing part **13** and a conveying part **12** engaged with the collecting part **11**; in this embodiment, the collecting part **11** is cylindrical, and the collecting part **11** has a collecting inlet **111** and a collecting outlet **112**, and the collecting part **11** gradually contracts from the collecting inlet **111** to the collecting outlet **112**, so that fruits can easily enter the conveying part **12** through the collecting part **11**; stabilizers **113** are embedded at the collecting inlet **111** and the collecting outlet **112**, and between the collecting inlet **111** and the collecting outlet **112**; in another embodiment, the collecting part **11** may also be a curved cylindrical part with an inlet end and an outlet end.

(21) The grabbing part **1313** is engaged at the end of the collecting inlet **111**; in this embodiment, the grabbing part **13** is seven strip-shaped parts **131** with bending parts **132**, and the bending angle α of the bending parts **132** is greater than or equal to 60 degrees and less than or equal to 90 degrees. In this embodiment, the bending angle α is 90 degrees. A sharp edge **1321** is formed on one side of the bending part **132**, and a user can cut the fruit handle through the sharp edge **1321** when using, so that the picking process is more labor-saving; the axis of the bending part **132**

intersects with the central axis of the collecting inlet **111**, and the strip-shaped part **131** is equidistantly engaged with the collecting inlet **111** of the collecting part **11** through the encapsulation process, and the end of the strip-shaped part **131** engaged with the collecting inlet **111** is fixedly connected with the stabilizer **113**, so that the connection between the strip-shaped part **131** and the collecting part **11** is more stable; in this embodiment, the strip-shaped part **131** is welded with the stabilizer **113**, and the strip-shaped part **131** can be wound around the stabilizer **113** in addition to welding; grabbing gaps **133** are formed between the strip-shaped parts **131**, and the width of the grabbing gaps **133** is smaller than that of the fruit.

(22) In another embodiment, the strip-shaped parts **131** can be detachably engaged at the collecting inlet **111** of the collecting part **11** through threads or buckles, and the distance between the strip-shaped parts **131** can be closer. By disassembling or installing the strip-shaped parts **131**, the width of the grabbing gap **133** can be adjusted, so that it can adapt to different fruit sizes.

(23) As shown in FIG. 3 and FIG. 5, in this embodiment, the conveying part **12** is a fabric bag, which has an input port end **121** and an output port end **122**. The input port end **121** of the conveying part **12** can be tied at the collecting outlet **112** of the collecting part **11** through a belt-like part, or can be detachably connected with the collecting outlet **112** of the collecting part **11** through a clamp; an elastic material is sewed in the conveying part **12**, so that the conveying part **12** can longitudinally extend and contract, avoiding the U-shaped structure of the conveying part **12** when in use, which will otherwise cause the fruits to be accumulated in the conveying part **12**; at the same time, the conveying part **12** is provided with a buffer part **123**, which is a soft brush in this embodiment; when the fruit falls in the conveying part **12**, the falling speed of the fruit is reduced by the soft brush to prevent the fruit from being broken during the falling process.

(24) As shown in FIG. 3 and FIG. 6, in this embodiment, the collecting part **11** is provided with a fifth connecting part **114**, and the fifth connecting part **114** is internally provided with a first connecting channel **2211**. The first connecting channel **2211** is internally provided with an internal thread, which is detachably engaged with the supporting part **2** through the fifth connecting part **114**. The supporting part **2** includes a first supporting body **21** and at least one second supporting body **22**, and the first supporting body **21** has a holding end **211** and a first connecting end **212**. The holding end **211** of the first supporting body **21** is provided with a plurality of grooves **2111**, and the bottoms of the grooves **2111** are at a distance from the surface of the holding end **211**. The holding end **211** has an uneven surface due to the grooves **2111**, which increases the friction between the supporting part **2** and the palm of the user, so that the supporting part **2** is not easy to get rid of when in use, and safety accidents are avoided. A hanging part **2112** is arranged at the port of the holding end **211**, and the hanging part **2112** is provided with a hanging hole **2113**, so that the supporting part **2** can be hung on the wall when placed.

(25) In this embodiment, the first connecting end **212** of the first supporting body **21** contracts inward to form a connecting part **213**; a first limiting inclined plane **214** is formed at the contraction position; the port of the connecting part **213** is provided with a first jack **215**, and the connector **23** is detachably connected through the first jack **215**.

(26) As shown in FIG. 7, the connector **23** has an inserting end **231** and a connecting end **232**; the end of the inserting end **231** contracts toward the central axis to form a guide inclined surface **2311**, which makes it easier for the connector **23** to be inserted into the first jack **215**; a first limiting part **233** is arranged between the inserting end **231** and the connecting end **232**; through the first limiting part **233**, only the inserting end **231** of the connector **23** is inserted into the first jack **215**, and the connecting end **232** is left outside the first jack **215**; the connecting end **232** has a first connector **2321**; the first connector **2321** is tapered and the top extends horizontally forward to form a first connector **2322**; the first connector **2321** is provided with external threads.

(27) As shown in FIGS. 8 and 9, the second supporting body **22** has a first inserting end **221** and a second connecting end **222**; the port of the first inserting end **221** is provided with a first connecting channel **2211** corresponding to the first connector **2321**, and an internal thread is

provided in the first connecting channel **2211**, so that the first supporting body **21** and the second supporting body **22** are connected through threads; the first connecting channel **2211** is internally provided with a second connecting channel **2212** corresponding to the first connector **2322**; when the first supporting body **21** is connected with the second supporting body **22**, the connecting end **232** of the connector **23** inserted in the first supporting body **21** is matched with the first connecting channel **2211** and the second connecting channel **2212** in the second supporting body **22**, so that the connection is more stable.

(28) The second connecting end **222** contracts inward to form a third connecting part **223**, a second limiting inclined plane **224** is formed at the contraction position; the port of the third connecting part **223** is provided with a second jack **225**, and the connector **23** is detachably connected through the second jack **225**; the fifth connecting part **114** of the collecting part **11** cooperates with the connector **23**, so that the second supporting body **22** can be detachably connected with the collecting part **11**.

(29) When the user needs to lengthen the supporting part **2**, the number of connected second supporting bodies **22** can be increased, and the connector **23** inserted through the third connecting part **223** of the second supporting body **22** is matched with the first inserting end **221** of another second supporting body **22**, so that the second supporting bodies **22** can be detachably connected; the user can correspondingly increase the number of connected second supporting bodies **22** according to the usage scenario.

(30) As shown in FIG. **10**, a connecting sleeve **24** is sleeved at the joint of the first supporting body **21** and the second supporting body **22**; a second limiting part **241** is arranged in the connecting sleeve **24**. When the connecting sleeve **24** is sleeved on the first supporting body **21** or the second supporting body **22**, the port of the first connecting end **212** or the port of the first inserting end **221** abuts against the second limiting part **241**; through the cooperation of the second limiting part **241** with the port of the first connecting end **212** and the port of the first inserting end **221**, when the first supporting body **21** is connected with the second supporting body **22**, the connecting sleeve **24** is fixedly sleeved at a predetermined position and cannot move up and down; and when the number of the second supporting bodies **22** is required, the joint between the second supporting bodies **22** is also sleeved with a connecting sleeve **24**; by sleeving the connecting sleeve **24**, the connection between the supporting bodies is more stable, and the situation of disconnection during use is avoided.

(31) In another embodiment, the first supporting body **21** and the second supporting body **22** can be detachably connected through a buckle structure; the second supporting body **22** and the collecting part **11** can also be detachably connected through a buckle structure; the supporting bodies can also be connected in a socket way, and a limit protrusion and a limit groove are arranged at the connection position, so that the supporting bodies are fixed at a predetermined position; the user can select the number of connected supporting bodies according to the usage scene.

(32) As shown in FIGS. **11** and **12**, the container part **3** is provided with a fixing part, so that the container part **3** can be fixed on a user; the fixing part is made of materials with strong toughness, including but not limited to braids; the fixing part includes a first fixing part **31** and a second fixing part **32**. In this embodiment, the first fixing part **31** is two belt-shaped parts with two ends fixed on the container wall, and a fixing hole **311** is formed between the first fixing part **31** and the container part **3**, so that a user can pass his arms through the fixing holes **311** to fix the container part **3** on the user.

(33) The second fixing part **32** includes a first fixture **321** and a second fixture **322**. One ends of the first fixture **321** and the second fixture **322** are fixedly connected with the container wall, and the other ends of the first fixture **321** and the second fixture **322** are respectively connected with a male buckle and a female buckle, so that the first fixture **321** and the second fixture **322** are detachably connected through the male buckle and the female buckle. In another embodiment, the first fixture **321** and the second fixture **322** can also be detachably connected by a hook and loop; through the

second fixing part 32, the container part 3 can be connected with the waist of the user; the user can more firmly carry the container part 3 on his back.

(34) The container part 3 is provided with a first opening 33 and a second opening 34. In this embodiment, the first opening 33 is located above the side wall of the container part 3, and a fourth connecting part 35 is arranged at the first opening 33. The fourth connecting part 35 can be arranged on the outer wall or the inner wall of the container part 3.

(35) In this embodiment, the periphery of the fourth connecting part 35 is provided with a fixing groove 351, and the output port end 122 can be fixed in the fixing groove 351 on the fourth connecting part 35 through a belt-like member, so that the conveying part 12 can be detachably connected with the container part 3; alternatively, the output port end 122 can be fixed to the fourth connecting part 35 by the cooperation between the clamp and the fixing groove 351.

(36) In another embodiment, connectors can be respectively engaged at the output port end 122 and the first opening 33, and the detachable connection between the conveying part 12 and the container part 3 can be realized through the cooperation of the connectors. The connector can be a zipper structure or a part with a snap fit.

(37) In this embodiment, the second opening 34 is located at the top of the container part 3. When the user carries the container part 3 on his back, he can observe whether the container is full or not through the second opening 34. When the picked fruit falls on the ground, it is convenient for users to put the fruit directly into the container part 3.

(38) In another embodiment, the first opening 33 can be arranged at the top of the container part 3, and the output port end 122 can be detachably connected at the first opening 33 through the fourth connecting part 35.

(39) In this embodiment, the container part 3 is provided with a third opening 37 below the side wall of the container, and a cover 38 is connected to the third opening 37 through a fixture 36, and the opening and closing of the third opening 37 can be controlled through the fixture 36; in this embodiment, the fixture 36 has a zipper structure; in another embodiment, the fixture 36 can also be a buckle structure or a hook and loop.

(40) In another embodiment, the third opening 37 can be located at the bottom of the container, and the bottom wall of the container is connected through the fixture 36, and the opening and closing of the third opening 37 can be controlled through the fixture 36.

(41) A method for picking fruits by using a device includes the following steps: providing the device, which includes: a picking module and a container part 3, wherein the picking module is detachably connected with the container part 3; the picking module comprises a picking part 1 and a supporting part 2, and the picking part 1 is detachably connected with the supporting part 2; the picking part 1 comprises a collecting part 11, a grabbing part 13 and a conveying part 12 which are engaged with the collecting part 11; the collecting part 11 is provided with a collecting inlet 111 end and a collecting outlet 112 end, the conveying part 12 is provided with an input port end 121 and an output port end 122, the grabbing part 13 is engaged at the collecting inlet 111 end, and the input port end 121 of the conveying part 12 is detachably connected with the collecting outlet 112 of the collecting part 11; the supporting part 2 comprises a first supporting body 21 and at least one second supporting body 22, and the first supporting body 21 and the second supporting body 22 are detachably connected; the first supporting body 21 has a holding end 211 and a first connecting end 212, one end of the second supporting body 22 is engaged with the connecting end 232 of the first supporting body 21, and the other end of the second supporting body 22 detachably connected with the picking part 1; the container part 3 is provided with a first opening 33 and a second opening 34, and the first opening 33 is detachably connected with the output port end 122 of the conveying part 12; the device is used for picking fruits; the fruits are grabbed by the grabbing part 13, and the fruit handle passes through the grabbing gap 133, so that the fruit is below the bending part 132 of the strip-shaped part 131; because the width of the fruit is larger than the grabbing gap 133, the fruit will be stuck below the bending part 132, and at this time, the user provides a downward force to

make the fruit fall off the fruit handle; fruits enter the collecting part **11** from the inlet end of the collecting part **11** and into the conveying part **12** through the outlet end of the collecting part **11**, and the fruits in the conveying part **12** enter the container part from the output port end **122** under the action of gravity.

(42) The terms “comprising,” “including,” “having,” and the like are synonymous and are used inclusively, in an open-ended fashion, and do not exclude additional elements, features, acts, operations, and so forth. Also, the term “or” is used in its inclusive sense (and not in its exclusive sense) so that when used, for example, to connect a list of elements, the term “or” means one, some, or all of the elements in the list. The use of “adapted to” or “configured to” herein is meant as open and inclusive language that does not foreclose devices adapted to or configured to perform additional tasks or steps. Additionally, the use of “based on” is meant to be open and inclusive, in that a process, step, calculation, or other action “based on” one or more recited conditions or values may, in practice, be based on additional conditions or values beyond those recited. Similarly, the use of “based at least in part on” is meant to be open and inclusive, in that a process, step, calculation, or other action “based at least in part on” one or more recited conditions or values may, in practice, be based on additional conditions or values beyond those recited. Headings, lists, and numbering included herein are for ease of explanation only and are not meant to be limiting.

(43) The various features and processes described above may be used independently of one another, or may be combined in various ways. All possible combinations and sub-combinations are intended to fall within the scope of the present disclosure. In addition, certain method or process blocks may be omitted in some implementations. The methods and processes described herein are also not limited to any particular sequence, and the blocks or states relating thereto can be performed in other sequences that are appropriate. For example, described blocks or states may be performed in an order other than that specifically disclosed, or multiple blocks or states may be combined in a single block or state. The example blocks or states may be performed in serial, in parallel, or in some other manner. Blocks or states may be added to or removed from the disclosed examples. Similarly, the example systems and components described herein may be configured differently than described. For example, elements may be added to, removed from, or rearranged compared to the disclosed examples.

(44) The invention has now been described in detail for the purposes of clarity and understanding. However, those skilled in the art will appreciate that certain changes and modifications may be practiced within the scope of the appended claims. Conditional language used herein, such as, among others, “can,” “could,” “might,” “may,” “e.g.,” and the like, unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain examples include, while other examples do not include, certain features, elements, and/or steps. Thus, such conditional language is not generally intended to imply that features, elements and/or steps are in any way required for one or more examples or that one or more examples necessarily include logic for deciding, with or without author input or prompting, whether these features, elements and/or steps are included or are to be performed in any particular example.

Claims

1. A portable fruit picker with an adjustable height, comprising a picking module and a container part (**3**), wherein said picking module is detachably connected with said container part (**3**); and wherein said picking module comprises a picking part (**1**) and a supporting part (**2**), and said picking part (**1**) is detachably connected with said supporting part (**2**); and wherein, said picking part (**1**) comprises a collecting part (**11**), a grabbing part (**13**) and a conveying part (**12**) which are engaged with said collecting part (**11**); and wherein, said collecting part (**11**) is provided with a collecting inlet (**111**) end and a collecting outlet (**112**) end, said conveying part (**12**) is provided with an input port end (**121**) and an output port end (**122**), said grabbing part (**13**) is engaged at said

collecting inlet (111) end, and said input port end (121) of said conveying part (12) is detachably connected with said collecting outlet (112) of said collecting part (11); and wherein, said supporting part (2) comprises a first supporting body (21) and at least one second supporting body (22), and said first supporting body (21) and said second supporting body (22) are detachably connected; and wherein, said first supporting body (21) has a holding end (211) and a first connecting end (212), one end of said second supporting body (22) is engaged with said first connecting end (212) of said first supporting body (21), and an opposite end of said second supporting body (22) detachably connected with said picking part (1); and wherein said second supporting body (22) has a second connecting end (222) and a first inserting end (221); and said second connecting end (222) contracts inward to form a third connecting part (223); and a second limiting inclined plane (224) is formed at a contraction position; and wherein a connecting sleeve (24) is sleeved at a joint of said first supporting body (21) and said second supporting body (22); and a second limiting part (241) is arranged in said connecting sleeve (24), so that said connecting sleeve (24) is fixedly sleeved at a joint of said first supporting body (21) and said second supporting body (22); and wherein said container part (3) is provided with a first opening (33) and a second opening (34), and said first opening (33) is detachably connected with said output port end (122) of said conveying part (12).

2. The portable fruit picker with an adjustable height according to claim 1, wherein said conveying part (12) has an elastic feature configured to longitudinally extend and contract, said conveying part (12); and a buffer part (123) is arranged in said conveying part (12) and configured such, that fruits pass through said conveying part (12) at a slow speed.

3. The portable fruit picker with an adjustable height according to claim 2, wherein said first connecting end (212) of said first supporting body (21) contracts inward to form a connecting part (213); and a first limiting inclined plane (214) is formed at a contraction position.

4. The portable fruit picker with an adjustable height according to claim 3, wherein said holding end (211) of said first supporting body (21) is provided with a plurality of grooves (2111), so that said holding end (211) has an uneven surface; and said connecting part (213) is detachably connected with a connector (23).

5. The portable fruit picker with an adjustable height according to claim 4, wherein said connector (23) has an inserting end (231) and a connecting end (232); and an end of said inserting end (231) contracts towards central axis to form a guide inclined plane (2311); and a first limiting part (233) is arranged between said inserting end (231) and said first connecting end (212).

6. The portable fruit picker with an adjustable height according to claim 5, wherein said connecting end (232) has a first connecting part (2321); and said first connecting part (2321) is tapered, and a top thereof horizontally extends forward to form a second connecting part (2322); and said first connecting part (2321) is provided with an external thread.

7. The portable fruit picker with an adjustable height according to claim 1, wherein a port of said first inserting end (221) is provided with a first connecting channel (2211) corresponding to a first connecting part (2321), and a second connecting channel (2212) corresponding to a second connecting part (2322) is arranged in said first connecting channel (2211); and said first connecting channel (2211) is internally provided with an internal thread, so that said first supporting body (21) and said second supporting body (22) are connected through threads.

8. The portable fruit picker with an adjustable height according to claim 1, wherein said container part (3) is provided with a fixing part, which comprises a first fixing part (31) and a second fixing part (32), and said second fixing part (32) comprises a first fixture (321) and a second fixture (322); and said first fixture (321) and said second fixture (322) are detachably connected.

9. The portable fruit picker with an adjustable height according to claim 8, wherein said container part (3) is provided with a third opening (37), and said third opening (37) is connected with a cover body (38) through a fixture (36), and opening and closing of said third opening (37) is configured to be controlled through said fixture (36).

10. The portable fruit picker with an adjustable height according to claim 8, wherein said first

opening (33) is provided with a fourth connecting part (35), and said output port end (122) is detachably connected with said first opening (33) through said fourth connecting part (35).
